



Modelling Financial Markets

Seminar 11: Continuous-Time Models — Hands-on



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Acronyms used in this material

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ FRED: Federal Reserve Economic Data
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ OLS: Ordinary Least Squares
- ▣ OU: Ornstein–Uhlenbeck (process)
- ▣ PDE: Partial Differential Equation
- ▣ RFSV: Rough Fractional Stochastic Volatility (model)
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Change measure with Girsanov, price bonds with Feynman–Kac and derive conditional moments of CIR and Heston
- Map between continuous-time models and their discrete-time versions (AR(1), Poisson mixtures)
- Estimate GBM, Vasicek, Merton and Heston parameters with standard errors
- Recognise when standard inference fails: biased $\hat{\kappa}$, boundary likelihood-ratio tests, unbounded likelihoods, noisy variance proxies

- Background: Lecture 11; textbooks Franke, Härdle & Hafner (2019) and Borak, Härdle & López-Cabrera (2013)

Structure

- **Part A:** nine derivations (A1–A9)
- **Part B:** nine computational exercises with inference (B1–B9)
- **Part C:** an open question, a seed for the team project (C1), the critique of an AI answer (C2) and a project seed from the lecture case study on rough volatility (C3)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- ▣ Time in years, daily step $\Delta t = 1/252$ (Bitcoin 1/365); log returns; drifts and volatilities per year
- ▣ Report every estimate with its standard error or confidence interval (CI)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Girsanov for GBM), A3 (Vasicek bond by Feynman–Kac), A5 (CIR moments and Feller), A7 (Merton cumulants and the information on λ), A9 (AR(1) to OU with the delta method)
- Part B: B1 (orders of convergence), B2 (GBM on the S&P 500), B4 (Vasicek and the bias of $\hat{\kappa}$), B6 (Merton and a bootstrap LR test)

Proposed: you solve them

- Part A: A2 (Siegel's paradox and the numéraire; model: A1), A4 (Itô integrals; model: A3), A6 (VIX² under Heston; model: A5), A8 (rare large jumps; model: A7)
- Part B: B3 (GBM on BET and Bitcoin; model: B2), B5 (OU for the VIX; model: B4), B7 (jumps in Bitcoin; model: B6), B8 (Heston from the VIX; model: B4, B6), B9 (nonparametric diffusion; model: B4)
- Part C: C1, open question and project seed; C2, critique of an AI answer; C3, is volatility rough? (lecture case study; model: B1, B4)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter11_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`
- Series: S&P 500 (1990–2026), BET (2000–2026), Bitcoin (2014–2026), VIX (1990–2026); daily closes up to 18 September 2026
- 3-month US Treasury bill rate (FRED series DTB3), 1954–2026
- Equity indices: weekdays only, stale quotes removed (unchanged close and no intraday range), genuine zero returns kept; Bitcoin: 7 days a week

Formulas Used Today

- ▣ Itô's lemma: $dX = a dt + b dW \Rightarrow df(X) = (af' + \frac{1}{2}b^2f'')dt + bf' dW$
- ▣ Girsanov: $dQ/dP = e^{-\theta W_T - \theta^2 T/2}$, $\tilde{W}_t = W_t + \theta t$; Feynman–Kac: $u_t + au_x + \frac{1}{2}b^2u_{xx} - ru = 0$
- ▣ GBM: $\ln(S_T/S_0) \sim N((\mu - \sigma^2/2)T, \sigma^2 T)$; $\mathbb{E}[S_T] = S_0 e^{\mu T}$
- ▣ OU: $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$; AR(1): $b = e^{-\kappa \Delta t}$
- ▣ Merton cumulants: $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t(\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t(\mu_J^3 + 3\mu_J \sigma_J^2)$, $\kappa_4 = \lambda \Delta t(\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$
- ▣ Heston: $dv = \kappa(\theta - v)dt + \xi\sqrt{v}dW^{(2)}$, $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$; Feller: $2\kappa\theta \geq \xi^2$

A1: GBM under P and Q (Girsanov) [Solved]

Task

- ▣ Context: a stock follows GBM $dS = \mu S dt + \sigma S dW$ under the real-world measure P , with $\mu = 8\%$ and $\sigma = 20\%$ per year **Girsanov (1960)**
- ▣ Assumptions: constant risk-free rate $r = 3\%$ per year, no dividends, horizon $T = 10$ years; the exercise uses given numbers, no market data
- ▣ (a) Find the market price of risk θ for which the discounted price $e^{-rt}S_t$ has zero drift, and write the Radon–Nikodym density $Z_T = dQ/dP$ on $\mathcal{F}_T$
- ▣ (b) Compute $\mathbb{E}^P[Z_T S_T]$ in closed form, show that it equals $S_0 e^{rT}$, and state whether σ changes under Q
- ▣ (c) Compute $P(S_{10} < S_0)$ under P and under Q from the lognormal law of S_T
- ▣ Report: θ , Z_T , $\mathbb{E}^Q[S_{10}]/S_0$ and the two probabilities; interpretation: which probability is a forecast and which one is a pricing weight?

A1: Solution (1/2) [Solved]

Solution

- Step 1: by Itô's product rule, $d(e^{-rt}S_t) = e^{-rt}S_t[(\mu - r)dt + \sigma dW_t]$
- Step 2: substitute $dW = d\tilde{W} - \theta dt$; the drift becomes $\mu - r - \sigma\theta$, which is zero for $\theta = (\mu - r)/\sigma = (0.08 - 0.03)/0.20 = 0.25$
- Step 3: Girsanov with a constant θ gives $Z_T = \exp(-\theta W_T - \frac{1}{2}\theta^2 T)$; the Novikov condition holds because θ is constant
- Step 4: $\mathbb{E}^P[Z_T S_T] = S_0 e^{(\mu - \sigma^2/2 - \theta^2/2)T} \mathbb{E}[e^{(\sigma - \theta)W_T}] = S_0 e^{(\mu - \sigma^2/2 - \theta^2/2 + (\sigma - \theta)^2/2)T} = S_0 e^{(\mu - \sigma\theta)T} = S_0 e^{rT}$
- Result: $\mathbb{E}^Q[S_{10}]/S_0 = e^{0.3} = 1.350$, the growth of a risk-free deposit

A1: Solution (2/2) [Solved]

Solution

- Step 5: under Q , $dS = rS dt + \sigma S d\tilde{W}$: σ is unchanged, because the quadratic variation $[\ln S]_T = \sigma^2 T$ is a property of the path, the same under equivalent measures
- Step 6: $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with the log drift $m_P = \mu - \sigma^2/2 = 6\%$ under P and $m_Q = r - \sigma^2/2 = 1\%$ under Q
- Under P : $\Phi(-0.06\sqrt{10}/0.20) = \Phi(-0.949) = 17.1\%$; under Q : $\Phi(-0.01\sqrt{10}/0.20) = \Phi(-0.158) = 43.7\%$
- Interpretation: the P -probability is the forecast of a loss; the Q -probability is a pricing weight that embeds the risk premium, so it overstates bad outcomes; the whole gap comes from the drift, since σ is common to both

A2: Siegel's Paradox and the Numéraire [Proposed]

Solution discussed in class

Task

- ▣ Context: the EUR/RON rate X_t (RON per euro) follows GBM with $\mu = 2\%$ and $\sigma = 4.4\%$ per year under P ; $Y_t = 1/X_t$ is the RON/EUR rate
- ▣ Assumptions: constant interest rates r_{RON} and r_{EUR} ; given numbers, no market data
- ▣ (a) Apply Itô's lemma to $f(x) = 1/x$ and write dY_t
- ▣ (b) Compute the expected percentage drift of Y with four decimals and compare it with $-\mu = -2\%$
- ▣ (c) Compute the log drifts of X and Y and compare them
- ▣ (d) Show that the forward $F = X_0 e^{(r_{RON} - r_{EUR})T}$ equals $\mathbb{E}^{Q^{RON}}[X_T]$, that $1/F = \mathbb{E}^{Q^{EUR}}[1/X_T]$, and write the density dQ^{EUR}/dQ^{RON}
- ▣ Model: A1
- ▣ Report: the drifts in (b)–(c) and the density in (d); interpretation: both a RON and a euro investor seem to expect a gain from the forward; why is there no arbitrage?

A3: The Vasicek Bond by Feynman–Kac [Solved]

Task

- ▣ Context: under the risk-neutral measure Q the short rate is Vasicek, $dr = \kappa(\theta - r)dt + \sigma d\tilde{W}$, with $\kappa = 0.5$, $\theta = 3\%$, $\sigma = 1\%$ per year and today $r_0 = 6\%$ [Vasicek \(1977\)](#)
- ▣ Assumptions: the market price of risk is already contained in θ ; a zero-coupon bond pays 1 at T ; given numbers, no market data
- ▣ (a) Use the Feynman–Kac theorem with payoff $g = 1$ and discount rate r to write the PDE and the terminal condition for $P(t, T) = \mathbb{E}^Q[e^{-\int_t^T r_s ds} \mid r_t]$ [Kac \(1949\)](#)
- ▣ (b) Insert the ansatz $P = \exp(A(\tau) - B(\tau)r)$, $\tau = T - t$, collect the terms in r and solve the two ordinary differential equations with $A(0) = B(0) = 0$
- ▣ (c) Compute B , A and the yield $y(\tau) = (Br_0 - A)/\tau$ for $\tau = 1, 5, 10$ years, and the limit y_∞
- ▣ Report: $B(5)$, $A(5)$, the three yields and y_∞ ; interpretation: why is the yield curve downward-sloping, and why is y_∞ below θ ?

A3: Solution (1/2) [Solved]

Solution

- (a) Feynman–Kac: $P_t + \kappa(\theta - r)P_r + \frac{1}{2}\sigma^2 P_{rr} - rP = 0$ with $P(T, T) = 1$
- (b) With $P = e^{A - Br}$: $P_t = (-A' + B'r)P$, $P_r = -BP$, $P_{rr} = B^2P$; dividing by P :
 $-A' + B'r - \kappa(\theta - r)B + \frac{1}{2}\sigma^2 B^2 - r = 0$ for every r
- Terms in r : $B' = 1 - \kappa B$, $B(0) = 0$, so $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$
- Constant terms: $A' = -\kappa\theta B + \frac{1}{2}\sigma^2 B^2$, $A(0) = 0$; integrating, $A(\tau) = \left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(B - \tau) - \frac{\sigma^2 B^2}{4\kappa}$

A3: Solution (2/2) [Solved]

Solution

- (c) $B(5) = (1 - e^{-2 \cdot 5})/0.5 = 1.836$; $A(5) = (0.03 - 0.0002)(1.836 - 5) - 0.0001 \cdot 1.836^2/2 = -0.0945$
- Yields: $y(1) = 5.36\%$, $y(5) = 4.09\%$, $y(10) = 3.58\%$; limit $y_\infty = \theta - \sigma^2/(2\kappa^2) = 0.03 - 0.0002 = 2.98\%$
- Interpretation: $r_0 = 6\%$ is above $\theta = 3\%$, so expected future rates fall and the curve slopes down
- $y_\infty < \theta$ because of convexity: $\mathbb{E}^Q[e^{-\int r}] > e^{-\mathbb{E}^Q[\int r]}$ (Jensen), so uncertainty about rates raises bond prices and lowers long yields by $\sigma^2/(2\kappa^2)$

A4: Itô Integrals [Proposed]

Solution discussed in class

Task

- Context: W is a standard Brownian motion on $[0, T]$; no data
- (a) Apply Itô's lemma to $f(x) = x^2$ and show that $\int_0^T W \, dW = \frac{1}{2}(W_T^2 - T)$
- (b) For $T = 1$ compute the mean and the variance of this integral, and check the variance with the Itô isometry
- (c) The Stratonovich integral equals $\frac{1}{2} W_T^2$: compute its mean for $T = 1$
- (d) Apply Itô's lemma to W_t^3 and deduce the mean and the variance of $\int_0^1 W^2 \, dW$
- Model: A3 (Itô's lemma applied to a function of the state)
- Report: the means and variances in (b)–(d); interpretation: why must an Itô integral with a square-integrable integrand have mean zero while the Stratonovich integral does not?

A5: CIR: Conditional Moments and the Feller Condition [Solved]

Task

- ▣ Context: the short rate follows CIR, $dr = \kappa(\theta - r)dt + \sigma\sqrt{r}dW$, with $\kappa = 0.11$, $\theta = 5.5\%$, $\sigma = 0.056$ per year, close to the exact CIR fit on daily DTB3, 1954–2007 (B4 (d)) [Cox, Ingersoll & Ross \(1985\)](#)
- ▣ Assumptions: today $r_0 = 3\%$; horizon $t = 1$ year
- ▣ (a) Apply Itô's lemma to $e^{\kappa t}r_t$, take expectations and derive $\mathbb{E}[r_t | r_0]$
- ▣ (b) Apply Itô's lemma to r_t^2 , write the ordinary differential equation for $m_2(t) = \mathbb{E}[r_t^2]$, solve it and derive $\text{Var}(r_t | r_0)$
- ▣ (c) State the transition law $2c r_t | r_0 \sim \chi_\nu^2(\lambda)$ and compute c , ν , λ and the 5% and 95% quantiles of r_1
- ▣ (d) Show that the Feller condition $2\kappa\theta \geq \sigma^2$ is equivalent to $\nu \geq 2$ and evaluate the Feller ratio
- ▣ Report: mean, standard deviation, c , ν , λ , the 90% interval and the Feller ratio; interpretation: why is the interval asymmetric around the mean, and can the rate become negative?

A5: Solution (1/2) [Solved]

Solution

- (a) $d(e^{\kappa t}r) = e^{\kappa t}(\kappa r dt + dr) = \kappa\theta e^{\kappa t} dt + \sigma e^{\kappa t}\sqrt{r} dW$; the integrand is square-integrable ($\mathbb{E}[r_s] < \infty$), so the stochastic integral has mean zero and $\mathbb{E}[r_t] = \theta + (r_0 - \theta)e^{-\kappa t}$
- (b) $d(r^2) = 2r dr + \sigma^2 r dt = [(2\kappa\theta + \sigma^2)r - 2\kappa r^2] dt + 2\sigma r^{3/2} dW$
- Taking expectations: $m'_2 = (2\kappa\theta + \sigma^2)\mathbb{E}[r_t] - 2\kappa m_2$; solving and subtracting $\mathbb{E}[r_t]^2$:

$$\text{Var} = r_0 \frac{\sigma^2}{\kappa} (e^{-\kappa t} - e^{-2\kappa t}) + \frac{\theta \sigma^2}{2\kappa} (1 - e^{-\kappa t})^2$$
- Numbers: $\mathbb{E}[r_1] = 0.055 + (0.03 - 0.055)e^{-0.11} = 3.26\%$; standard deviation 0.94%

A5: Solution (2/2) [Solved]

Solution

- (c) $c = \frac{2\kappa}{\sigma^2(1-e^{-\kappa t})} = 673$, $\nu = \frac{4\kappa\theta}{\sigma^2} = 7.72$, $\lambda = 2c r_0 e^{-\kappa t} = 36.2$
- Quantiles of $\chi^2_\nu(\lambda)$ divided by $2c$: 90% interval for r_1 is [1.85%, 4.92%]
- (d) $\nu = 2 \cdot (2\kappa\theta/\sigma^2)$, so $2\kappa\theta \geq \sigma^2 \Leftrightarrow \nu \geq 2$; Feller ratio $2\kappa\theta/\sigma^2 = 3.86$
- Interpretation: the noncentral χ^2 law is skewed to the right, because volatility $\sigma\sqrt{r}$ grows with the level: the interval reaches further above the mean than below it
- With $\nu \geq 2$ zero is never reached, so the rate stays strictly positive, unlike Vasicek; if Feller fails, zero is attainable but negative values never are: the 8 negative DTB3 quotes after 2008 lie outside CIR

A6: The VIX under Heston [Proposed]

Solution discussed in class

Task

- ▣ Context: under Q the Heston variance follows $dv = \kappa^Q(\theta^Q - v)dt + \xi\sqrt{v}d\tilde{W}^{(2)}$; the squared VIX is $y_t = \text{VIX}_t^2/100^2 = \frac{1}{\tau}\mathbb{E}^Q\left[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t\right]$, $\tau = 30/365$
- ▣ Given numbers: $\kappa^Q = 5$, $\theta^Q = 0.044$; the lecture CIR regression on y_t gives $\hat{\xi} = 0.56$
- ▣ (a) Using A5 (a) under Q , write $\mathbb{E}^Q[v_s \mid v_t]$, integrate over $[t, t + \tau]$ and show that $y_t = a + b v_t$; give a and b
- ▣ (b) Compute $\kappa^Q\tau$, b and a for the given numbers
- ▣ (c) From $dy = b dv$, express the diffusion of y through y , relate the estimated ξ_y to ξ at $v_t = \theta^Q$ and compute the corrected ξ
- ▣ (d) State which parameters (κ, θ, ρ) the regression on y_t recovers, and under which measure
- ▣ Model: A5
- ▣ Report: a , b and the corrected ξ ; interpretation: is $\hat{\xi} = 0.56$ too high or too low, and what else is needed to estimate κ^Q ?

A7: Merton: Cumulants and the Information on λ [Solved]

Task

- ▣ Context: Merton model with $\sigma = 15\%$, $\lambda = 5$ jumps per year, $\mu_J = -3\%$, $\sigma_J = 4\%$, daily step $\Delta t = 1/252$ Merton (1976)
- ▣ For (c): the S&P 500 sample spans $T = 36.7$ years; the lecture MLE is $\hat{\lambda} = 102$ with s.e. 11
- ▣ (a) Write the cumulant generating function $K(u)$ of the daily log return, derive $\kappa_k = \lambda \Delta t \mathbb{E}[Y^k]$ for $k \geq 3$, and compute κ_2 split into its diffusion and jump parts
- ▣ (b) Compute κ_3 , κ_4 , the skewness $\kappa_3/\kappa_2^{3/2}$ and the excess kurtosis κ_4/κ_2^2
- ▣ (c) Assume every jump is observed: write the Poisson log-likelihood of N_T , derive the Fisher information for λ and the s.e. of $\hat{\lambda} = N_T/T$ at $\lambda = 5$ and at $\lambda = 101$
- ▣ Report: κ_2 to κ_4 , skewness, excess kurtosis and the two standard errors; interpretation: why is the MLE s.e. much larger than the s.e. with observed jumps?

A7: Solution (1/2) [Solved]

Solution

- (a) $K(u) = u m \Delta t + \frac{1}{2} u^2 \sigma^2 \Delta t + \lambda \Delta t (\mathbb{E}[e^{uY}] - 1)$; the k -th derivative at 0 gives $\kappa_k = \lambda \Delta t \mathbb{E}[Y^k]$ for $k \geq 3$
- $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2) = 0.893 \times 10^{-4} + 0.496 \times 10^{-4} = 1.389 \times 10^{-4}$; jumps carry 35.7% of the variance
- (b) $\mathbb{E}[Y^3] = \mu_J^3 + 3\mu_J \sigma_J^2$, $\mathbb{E}[Y^4] = \mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4$: $\kappa_3 = -3.393 \times 10^{-6}$, $\kappa_4 = 3.399 \times 10^{-7}$
- Skewness -2.07 , excess kurtosis 17.6 : five modest jumps a year make daily returns strongly non-Normal

A7: Solution (2/2) [Solved]

Solution

- (c) $\ell(\lambda) = N_T \ln \lambda - \lambda T + \text{const}$; $\ell'' = -N_T/\lambda^2$, so $I(\lambda) = \mathbb{E}[N_T]/\lambda^2 = T/\lambda$ and $\text{s.e.}(\hat{\lambda}) = \sqrt{\lambda/T}$
- At $\lambda = 5$, $T = 36.7$: $\text{s.e.} \sqrt{5/36.7} = 0.37$, i.e. 7.4% of λ
- At $\hat{\lambda} = 102$, observed jumps would give $\text{s.e.} 1.67$; the MLE s.e. is 6.3 times larger
- Interpretation: the jumps are latent; the Poisson mixture must separate them from diffusive moves, which destroys most of the information about λ

A8: Rare, Large Jumps [Proposed]

Solution discussed in class

Task

- ▣ Context: Merton model with $\sigma = 15\%$ but rare crashes: $\lambda = 0.5$ per year, $\mu_J = -10\%$, $\sigma_J = 5\%$, daily step $\Delta t = 1/252$; sample span $T = 36.7$ years
- ▣ (a) Compute the daily skewness and excess kurtosis with the cumulant formulas of A7
- ▣ (b) Compute the probability of observing no jump in 1 year and in 10 years
- ▣ (c) Even with every crash observed, compute the s.e. of $\hat{\lambda}$ over 36.7 years and the number of years needed for a 20% relative s.e.
- ▣ (d) Explain what the lecture MLE ($\hat{\lambda} \approx 100$) captures instead
- ▣ Model: A7
- ▣ Report: skewness, excess kurtosis, the two probabilities, the s.e. and the years; interpretation: can crash risk be estimated from the history of one index?

A9: From an AR(1) to an OU Process, with Standard Errors [Solved]

Task

- ▣ Context: an AR(1) estimated on $n = 600$ months (50 years) of a rate in decimals, step $\Delta = 1/12$:
 $x_{t+1} = 0.0012 + 0.97 x_t + \varepsilon_t$, residual s.d. $\hat{s}_\varepsilon = 0.0025$
- ▣ Assumptions: the rate is an exactly sampled OU process, so $b = e^{-\kappa\Delta}$, $a = \theta(1 - b)$, $s^2 = \sigma^2(1 - b^2)/(2\kappa)$
- ▣ (a) Invert the three relations to obtain κ , θ , σ and the half-life $\ln 2/\kappa$
- ▣ (b) Use $\text{Var}(\hat{b}) \approx (1 - b^2)/n$ and the delta method to compute the s.e. of $\hat{\kappa}$ and of the half-life
- ▣ (c) Keep $T = 50$ years but sample daily ($\Delta = 1/252$): compute b , the s.e. of $\hat{\kappa}$ and the limit $\sqrt{2\kappa/T}$
- ▣ Report: κ , θ , σ , half-life with s.e. and the daily s.e.; interpretation: does daily sampling make κ more precise?

A9: Solution (1/2) [Solved]

Solution

- (a) $\kappa = -\ln b/\Delta = -12 \ln 0.97 = 0.366$; $\theta = a/(1 - b) = 0.0012/0.03 = 4.0\%$
- $\sigma = s\sqrt{2\kappa/(1 - b^2)} = 0.0025\sqrt{2 \cdot 0.366/(1 - 0.97^2)} = 0.88\%$; half-life $\ln 2/0.366 = 1.90$ years
- (b) $\text{s.e.}(\hat{b}) = \sqrt{(1 - 0.97^2)/600} = 0.0099$; $\partial\kappa/\partial b = -1/(b\Delta)$, so $\text{s.e.}(\hat{\kappa}) = 0.0099/(0.97/12) = 0.123$
- Half-life: derivative $-\ln 2/\kappa^2$, so $\text{s.e.} = (\ln 2/\kappa^2) \text{s.e.}(\hat{\kappa}) = 0.64$ years: a 95% CI of roughly ± 1.2 years

A9: Solution (2/2) [Solved]

Solution

- (c) Daily: $b = e^{-\kappa/252} = 0.99855$ and $n = 12\,600$; the same formulas give s.e. 0.121
- In general $\text{Var}(\hat{\kappa}) \approx (1 - b^2)/(nb^2\Delta^2) = (e^{2\kappa\Delta} - 1)/(T\Delta)$; as $\Delta \rightarrow 0$ it tends to $2\kappa/T$; limit $\sqrt{2\kappa/T} = 0.121$
- Interpretation: 21 times more observations give the same precision: κ is a long-span parameter, identified by calendar time, not by the number of points (lecture, Section 7)

B1: Orders of Convergence [Solved]

Task

- ▣ Setting: test equation $dX = 2X dt + X dW$, $X_0 = 1$, $T = 1$, with exact solution $X_T = \exp(1.5 + W_T)$ [Higham \(2001\)](#); 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to $\Delta t = 2^{-11}, \dots, 2^{-5}$ (common random numbers)
- ▣ (a) For each Δt compute the strong error $\mathbb{E}|X_T - X_T^{\Delta t}|$ of Euler–Maruyama and Milstein, averaged over the 20,000 paths
- ▣ (b) Regress $\ln(\text{error})$ on $\ln \Delta t$ by OLS; since all step sizes use the same paths, obtain the s.e. and a 95% CI of each slope by resampling whole paths (1,000 bootstrap samples, all step sizes together)
- ▣ (c) Compute the weak error $|\mathbb{E}[X_T^{\Delta t}] - \mathbb{E}[X_T]|$ of Euler–Maruyama exactly, from $\mathbb{E}[X_T^{\Delta t}] = (1 + 2\Delta t)^{1/\Delta t}$, and by Monte Carlo; estimate both slopes
- ▣ Report: a table of the four slopes with bootstrap s.e. and CI (none for the deterministic exact-mean slope)

Interpretation

- ▣ Are the slopes consistent with the theoretical orders 1/2 and 1? Why does the Monte Carlo weak error give a wrong slope, and how many paths would resolve it?

B1: Results [Solved]

	Slope	s.e.	95% CI
Strong, Euler–Maruyama	0.518	0.003	[0.512, 0.525]
Strong, Milstein	0.986	0.001	[0.984, 0.987]
Weak, Euler–Maruyama, exact mean	0.985	–	deterministic
Weak, Euler–Maruyama, Monte Carlo	0.511	0.269	[0.247, 1.289]

- Step 1: strong slopes are close to 0.5 and 1; halving Δt halves the Milstein error but cuts the Euler error only by $1 - 2^{-1/2} = 29\%$
- Step 2: the path-bootstrap CIs measure Monte Carlo noise only; they can exclude 1/2 and 1 because the errors at coarse steps are pre-asymptotic: the local slopes between the two finest steps are 0.492 (Euler) and 0.998 (Milstein)

 MFM_ch11_convergence

B1: Reading the Weak Error [Solved]

- Step 3: the exact weak slope, a deterministic number with no sampling CI, is close to 1, as theory predicts (local slope at the finest steps 0.998)
- Step 4: the s.d. of X_T is $e^2\sqrt{e-1} = 9.69$, so the Monte Carlo s.e. of the mean is $9.69/\sqrt{20\,000} = 0.068$
- Step 5: at $\Delta t = 2^{-11}$ the weak error is 0.0072, ten times smaller than this s.e., so the Monte Carlo curve flattens and its slope is wrong
- Step 6: to resolve it at the 95% level, $(1.96 \cdot 9.69/0.0072)^2 \approx 6\,937\,381$ paths are needed
- Interpretation: for expected values the Monte Carlo sample size, not the time step, is usually the binding constraint

 MFM_ch11_convergence

B2: GBM on the S&P 500 [Solved]

Task

- Data: S&P 500 daily closes from `data/market/GSPC.INDX.csv`, 2 January 1990 – 18 September 2026, weekdays without stale quotes; log returns r_t , $\Delta t = 1/252$, $n = 9\,245$, $T = 36.7$ years
- (a) Compute the MLE $\hat{\sigma}^2 = \sum (r_t - \bar{r})^2 / (n\Delta t)$, $\hat{m} = \bar{r} / \Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2 / 2$, with $\text{s.e.}(\hat{\sigma}) = \hat{\sigma} / \sqrt{2n}$, $\text{s.e.}(\hat{m}) = \hat{\sigma} / \sqrt{T}$ and a 95% CI for m
- (b) Test normality with Jarque–Bera and independence with Ljung–Box on $|r_t|$ (20 lags), both at 5%
- (c) Compute the variance ratio of non-overlapping 20-day sums to 20 times the daily variance (GBM: 1)
- (d) Compute the years of data needed for a 95% CI of ± 1 percentage point on m
- (e) The s.e. in (a) assume GBM; recompute them by a moving-block bootstrap (blocks of 63 days, 999 samples) [Künsch \(1989\)](#), and give $\hat{\sigma} \sqrt{(k+2)/(4n)}$, the s.e. of $\hat{\sigma}$ for independent returns with excess kurtosis k
- Report: estimates with GBM and block-bootstrap s.e., test statistics with critical values and decisions

Interpretation

- Which parameter is known well and which badly, and why? Which GBM assumption does each test reject? Can you still use $\hat{\sigma}$ for option pricing?

B2: Results [Solved]

- ▣ Step 1 (GBM s.e.): $\hat{\sigma} = 18.0\%$ (s.e. 0.13), $\hat{m} = 8.3\%$ (s.e. 3.0), $\hat{\mu} = 8.3\% + \hat{\sigma}^2/2 = 10.0\%$
- ▣ Step 2: GBM 95% CI for m : [2.5%, 14.2%]; ± 1 point would need $(1.96\hat{\sigma}/0.01)^2 = 1\,243$ years
- ▣ Step 3 (e): s.e. of $\hat{\sigma}$: 0.34 with kurtosis, 1.15 by block bootstrap, 8.7 times the GBM value; m : block s.e. 2.5, CI [3.5%, 13.3%]
- ▣ Step 4: Jarque–Bera 45 805 against the 5% critical value 5.99 (excess kurtosis 10.9): $p < 0.001$, Normality rejected
- ▣ Step 5: Ljung–Box on $|r_t|$: 13 381 against $\chi^2_{20,0.95} = 31.4$, lag-1 ACF 0.27: $p < 0.001$, independence rejected
- ▣ Step 6: variance ratio 0.75, below 1: mild mean reversion of daily moves
- ▣ Interpretation: σ is estimated far more precisely than the drift even without GBM; $\hat{\sigma}$ is an **average** of a time-varying volatility [Cont \(2001\)](#): a benchmark for pricing, which needs the risk-neutral dynamics of variance, not a calibration

B3: GBM on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- Data: BET daily closes from `data/market/BET.csv`, 2000–2026, weekdays without stale quotes, $\Delta t = 1/252$; Bitcoin from `data/market/BTC-USD.CC.csv`, 17 September 2014 – 18 September 2026, 7 days a week, $\Delta t = 1/365$
- (a) Repeat B2 (a), (d) and (e) for each market: $\hat{\sigma}$, $\hat{m}$ with GBM and block-bootstrap 95% CIs and the years needed for ± 1 point
- (b) Repeat B2 (b) and (c): Jarque–Bera, Ljung–Box on $|r_t|$ with 20 lags at 5%, and the 20-day variance ratio
- Model: B2
- Report: one table with both markets

Interpretation

- Is the drift of Bitcoin significantly positive? Which market is further from GBM, and in which dimension? What does a variance ratio above 1 for BET suggest?

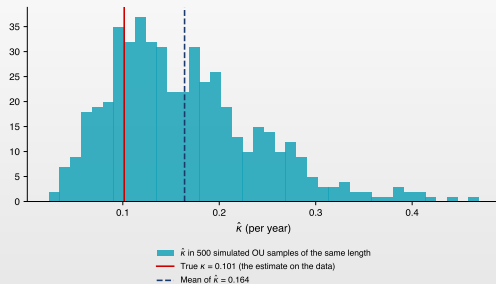
B4: Vasicek and the Bias of $\hat{\kappa}$ [Solved]

Task

- ▣ Data: 3-month US Treasury bill rate (FRED series DTB3), daily, 4 January 1954 – 18 September 2026, in decimals; $\Delta t = 1/252$, $n = 18\,170$ days (72.1 years)
- ▣ (a) Regress x_{t+1} on x_t by OLS (exact AR(1) of the OU process), invert to κ , θ , σ , compute delta-method s.e., a 95% CI for κ and the half-life [Vasicek \(1977\)](#)
- ▣ (b) Simulate 500 exact OU paths of the same length, started from the stationary law, with the estimates as the truth; re-estimate κ on each
- ▣ (c) Compute the bias (mean of the 500 estimates minus the truth), compare it with $4/T$ [Tang & Chen \(2009\)](#), and give the bias-corrected κ and half-life
- ▣ (d) Test $\kappa = 0$ with a simulated null: 500 random walks of the same length from x_0 , residual s.d. as estimated; p = share of $\hat{\kappa}$ at or above the observed one
- ▣ Report: estimates with s.e., the simulated distribution of $\hat{\kappa}$ and the corrected values

Interpretation

- ▣ Is mean reversion significant before and after the bias correction? What does the bias do to pricing and hedging with Vasicek?

B4: Simulated Bias of $\hat{\kappa}$ [Solved]

- 500 exact OU samples of 72.1 years with $\kappa = 0.101$: the histogram lies mostly to the right of the truth

B4: Results [Solved]

- Step 1: $\hat{b} = 0.99960$, so $\hat{\kappa} = -\ln \hat{b} / \Delta t = 0.101$ (s.e. 0.051), 95% CI [0.001, 0.202]
- Step 2: $\hat{\theta} = 4.55\%$ (s.e. 1.55), $\hat{\sigma} = 1.33\%$, half-life $\ln 2 / \hat{\kappa} = 6.8$ years
- Step 3: simulated mean $\hat{\kappa} = 0.164$ (s.d. 0.078): bias 0.063, close to $4/T = 0.055$; 78% of the estimates exceed the truth
- Step 4: bias-corrected $\kappa = 0.101 - 0.063 = 0.039$ (approximate 95% interval $0.039 \pm 1.96 \cdot 0.078 = [-0.114, 0.192]$), half-life 17.8 years instead of 6.8
- Step 5 (d): under a random walk, $\hat{\kappa}$ has 95% quantile 0.213; $p = 0.287$ for the observed 0.101
- Interpretation: the Wald CI barely excludes zero and ignores the bias; neither the calibrated test in (d) nor the corrected interval rejects $\kappa = 0$ at 5%. The corrected half-life is long, yet over 72 years the point estimate would shrink a deviation to 0.06 of its size: the data are compatible with slow reversion, but cannot exclude a random walk. Under a constant market price of risk, $\kappa^Q = \kappa$, so the upward bias makes Vasicek understate long-maturity rate risk

B4 (d): Exact CIR, Euler and CKLS [Solved]

Task

- Data: DTB3 on 4 January 1954 – 31 December 2007 only (after 2008 the series has 13 exact zeros and 8 negative quotes, outside the domain of the CIR and CKLS likelihoods, and a near-zero regime), daily ($n = 13\,487$, $\Delta = 1/252$) and month-end values ($n = 647$, $\Delta = 1/12$)
- (d1) Maximise the exact CIR likelihood ($2c\,r_{t+\Delta} \mid r_t \sim \chi^2_\nu(\lambda)$, A5) and the Euler (Gaussian) pseudo-likelihood; compare $\hat{\kappa}$ and $\hat{\theta}$ with Hessian s.e. [Cox, Ingersoll & Ross \(1985\)](#)
- (d2) Fit CKLS, $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$, by the Euler quasi-likelihood with γ free; compute the Hessian, sandwich and HAC sandwich (Newey–West, Bartlett kernel, $\lfloor 4(n/100)^{2/9} \rfloor$ lags) s.e. of $\hat{\gamma}$ [Chan, Karolyi, Longstaff & Sanders \(1992\)](#) [Newey & West \(1987\)](#)
- (d3) Test $\gamma = 0$ (Vasicek) and $\gamma = 1/2$ (CIR) by the Gaussian LR and by the HAC Wald test at 5% (critical value $\chi^2_{1,0.95} = 3.84$), at both frequencies
- Report: a table of $\hat{\kappa}$ (s.e.) and $\hat{\theta}$ by model and frequency; $\hat{\gamma}$ with both s.e.; the four statistics per frequency

Interpretation

- Which volatility specification do the data prefer, and does the answer change with the sampling frequency? Which test do you trust, LR or HAC Wald?

B4 (d): Estimates [Solved]

Model	Daily		Month-end	
	$\hat{\kappa}$ (s.e.)	$\hat{\theta}$ (%)	$\hat{\kappa}$ (s.e.)	$\hat{\theta}$ (%)
Vasicek, exact	0.158 (0.073)	5.39	0.201 (0.083)	5.36
CIR, exact	0.113 (0.060)	5.48	0.153 (0.070)	5.43
CIR, Euler	0.093 (0.060)	5.55	0.144 (0.069)	5.45
CKLS, Euler	0.080 (0.057)	5.84	0.144 (0.068)	5.46

- Step 1: exact and Euler CIR differ by less than one s.e. of $\hat{\kappa}$ at both frequencies: the discretisation error is small next to the long-span uncertainty
- Step 2: all $\hat{\kappa}$ lie between 0.08 and 0.20 with s.e. 0.06–0.08: none is estimated precisely, whatever the likelihood

B4 (d): Tests on γ [Solved]

- Step 3: $\hat{\gamma} = 0.77$ daily (s.e.: Hessian 0.008, sandwich 0.020, HAC 0.033 with 11 lags); $\hat{\gamma} = 0.60$ monthly (Hessian 0.031, sandwich 0.134, HAC 0.149 with 6 lags)
- Step 4, $\gamma = 0$: LR 7926.6 daily and 285.6 monthly; HAC Wald 545.1 and 16.1, all above 3.84: Vasicek is rejected at 5% at both frequencies
- Step 5, $\gamma = 1/2$: daily LR 1139.34 and HAC Wald 67.01 reject CIR (22.13 with 55 lags); monthly LR 10.07 rejects, but the HAC Wald $0.45 < 3.84$ does not
- Interpretation: volatility rises with the level (γ between $1/2$ and 1); omitted volatility clustering makes the scores heteroskedastic and serially correlated: the Hessian s.e. are far too small and the LR over-rejects; the contemporaneous sandwich fixes only heteroskedasticity, so trust the HAC Wald test
- Chan, Karolyi, Longstaff & Sanders (1992) found $\hat{\gamma} \approx 1.5$ on 1964–1989: the elasticity depends on the sample and on how volatility clustering is handled

B5: Mean Reversion of the VIX [Proposed]

Solution discussed in class

Task

- ▣ Data: VIX daily closes from data/market/VIX.INDX.csv, 1990–2026, weekdays; $x_t = \ln VIX_t$, $\Delta t = 1/252$, $n = 9\,283$
- ▣ (a) Fit the OU process by the exact AR(1) as in B4 (a); convert $\hat{\kappa}$ into a half-life in trading days, $252 \ln 2 / \hat{\kappa}$, with a 95% CI from the CI of κ
- ▣ (b) Estimate the bias of $\hat{\kappa}$ with 300 simulated OU paths of the same length
- ▣ (c) Compute the sample ACF of x_t at lags 1, 5, 20, 60, 120 and compare it with the OU prediction $\hat{b}^k$
- ▣ Model: B4
- ▣ Report: $\hat{\kappa}$, the long-run level $e^{\hat{\theta}}$, the half-life with CI, the bias and the ACF table

Interpretation

- ▣ Why is the bias much less important here than for interest rates? What does the ACF pattern say about a one-factor OU model?

B6: Merton and a Bootstrap Likelihood-Ratio Test [Solved]

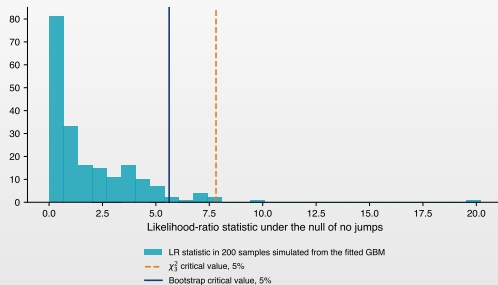
Task

- Data: S&P 500 daily closes from `data/market/GSPC.INDX.csv`, 2 January 1990 – 18 September 2026, weekdays without stale quotes; log returns r_t , $\Delta t = 1/252$, $n = 9\,245$, $T = 36.7$ years
- (a) Fit Merton ($m, \sigma, \lambda, \mu_J, \sigma_J$) by maximum likelihood on the Poisson mixture truncated at 12 jumps a day, over the bounded space $\sigma \geq 5\%$ (see (d)), from six starting points, with Hessian s.e. [Merton \(1976\)](#)
- (b) Compute the likelihood-ratio statistic against GBM ($\lambda = 0$) on this restricted space
- (c) Obtain its null distribution by parametric bootstrap: 200 GBM samples of the same length, both models refitted with the same restriction and the same six starts; give the 95% critical value, the number of exceedances and $p = (1 + \#\{LR^* \geq LR\})/201$
- (d) Compute the profile log-likelihood with σ fixed at 5%, 2%, 1%, the MLE under $\sigma \geq 5\%$, and decide whether the likelihood is bounded [Honoré \(1998\)](#)
- Report: parameters with s.e., LR, bootstrap and χ^2_3 critical values, p -value, the profile table

Interpretation

- Why is χ^2_3 not valid here? Does the fitted model match the kurtosis and skewness of the data? Is the likelihood bounded, and what does the reported MLE then mean?

B6: Bootstrap Null Distribution [Solved]



- LR statistics in 200 GBM samples, refitted by both models, with the bootstrap and χ^2_3 critical values at 5%

B6: Results [Solved]

- Step 1: $\hat{\sigma} = 9.44\%$ (0.32), $\hat{\lambda} = 101.9$ (10.5) per year, $\hat{\mu}_J = -0.162\%$ (0.034), $\hat{\sigma}_J = 1.44\%$ (0.06)
- Step 2: implied skewness -0.31 (data -0.36), excess kurtosis 3.71 (data 10.9): two thirds of the kurtosis is missed
- Step 3: $LR = 2401$ ($\sigma \geq 5\%$, not binding); bootstrap 95% critical value 5.62 against $\chi^2_{3,0.95} = 7.81$; 0 of the 200 bootstrap values exceed the LR, so $\hat{p} = (1 + 0)/201 = 0.0050$ (95% Monte Carlo upper bound 0.015): GBM is rejected at 5%
- Interpretation: under H_0 , $\lambda = 0$ is on the boundary and μ_J , σ_J are unidentified, so Wilks' theorem fails; jumps are supported, but the i.i.d. jump model is not the right description of the tails

B6 (d): Profile Likelihood [Solved]

σ (fixed)	log-likelihood	$\hat{\lambda}$ per year	$\hat{\sigma}_J$ (%)
9.4% (MLE)	29 500	101.9	1.44
5%	29 422	302	0.93
2%	29 265	555	0.70
1%	29 165	710	0.62

- Step 4: the profile falls as σ shrinks while $\hat{\lambda}$ grows; the MLE $\hat{\sigma} = 9.4\%$ satisfies $\sigma \geq 5\%$, so the constraint does not bind
- Step 5: yet $\sup = +\infty$: set $m\Delta t$ equal to one observed return; its term $\ln(e^{-\lambda\Delta t}\varphi(0; 0, \sigma^2\Delta t))$ equals 6.0, 10.7, 19.9, 38.3 at $\sigma = 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}$, while the other returns keep finite densities from the jump components
- Interpretation: as for Normal mixtures, the MLE is an interior local maximum; the global LR against GBM would be infinite, so the test in (b)–(c) is defined on $\sigma \geq 5\%$, with the same rule for data and bootstrap [Honoré \(1998\)](#)

B7: Jumps in Bitcoin [Proposed]

Solution discussed in class

Task

- Data: Bitcoin daily closes from `data/market/BTC-USD.CC.csv`, 17 September 2014 – 18 September 2026, 7 days a week; log returns, $\Delta t = 1/365$
- (a) Fit Merton by maximum likelihood as in B6 (a); report the parameters with s.e., the implied excess kurtosis and the jump share of variance
- (b) Apply the Lee–Mykland test at $\alpha = 1\%$ with a bipower window of $K = 16$ days [Lee & Mykland \(2008\)](#); count the jumps per year and their mean absolute size
- Model: B6
- Report: the Merton estimates and the Lee–Mykland counts side by side

Interpretation

- Why do $\hat{\lambda}$ and the Lee–Mykland count differ by two orders of magnitude? Which estimate would you use to price a one-week Bitcoin option?

B8: Heston Parameters from the VIX [Proposed]

Solution discussed in class

Task

- Data: S&P 500 (as in B2) and VIX (data/market/VIX.INDX.csv) on common days, 1990–2026; $y_t = (\text{VIX}_t/100)^2$ in place of v_t , $\Delta t = 1/252$ [Heston \(1993\)](#)
- (a) Run the discretised CIR regression $(y_{t+\Delta} - y_t)/\sqrt{y_t} = \kappa\theta\Delta t/\sqrt{y_t} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\varepsilon_t$; give κ , θ , ξ , the OLS s.e. and a 95% CI for κ
- (b) Estimate ρ as the correlation of return and variance shocks, with a Fisher-z 95% CI; compute the Feller ratio $2\kappa\theta/\xi^2$
- (c) Repeat (a)–(b) on 1990–2007 and on 2008–2026
- Model: B4, B6
- Report: a table with the full sample and the two halves

Interpretation

- Why is $\sqrt{\theta}$ from the VIX above the realised volatility? Are the parameters stable, and is the Feller condition met in both halves?

B8 (d)–(e): The Affine Map and Realised Variance [Proposed]

Solution discussed in class

Task

- Data: as in B8, plus SPY 5-minute realised variance including the overnight return (Chapter 9, Quantlets/Ch_09/ch9_rv_spy.csv), 12 October 2020 – 18 September 2026, annualised with $\Delta t = 1/252$
- (d) Correct $\hat{\xi}$ for the affine map $y = a + b v$ of A6 with $\kappa^Q = 5$ and state which estimates need no correction
- (e) On the common days, estimate ρ twice, once with y_t and once with realised variance in place of v_t , each with a Fisher-z 95% CI
- Model: A6, B8
- Report: the corrected ξ and the two correlations with CIs

Interpretation

- Why is ρ from realised variance much closer to zero than ρ from the VIX, and which one would you use in a Heston model?

B9: Nonparametric Diffusion of the T-Bill Rate [Proposed]

Solution discussed in class

Task

- Data: DTB3 daily, 4 January 1954 – 31 December 2007, $\Delta = 1/252$; the parametric fits of B4 (d) [Bandi & Phillips \(2003\)](#)
- (a) On a grid of 60 levels between the 2% and 98% quantiles of r , compute Nadaraya–Watson estimates of $\mathbb{E}[\Delta r \mid r]/\Delta = a(r) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r]/\Delta = b^2(r) + O(\Delta)$ (drift and diffusion as $\Delta \rightarrow 0$) with a Gaussian kernel and $h = 1.06 s_r n^{-1/5}$, plus 95% heteroskedasticity-robust pointwise bands
- (b) Overlay the Vasicek, CIR and CKLS diffusions of B4 (d) and compute the share of levels at which each lies inside the band
- (c) Repeat with $h/2$ and $2h$; also report the log-slope of $\hat{b}(r)$ and the share of levels where the drift band contains zero
- Model: B4
- Report: the chart of the lecture (Section 7) and a table by bandwidth

Interpretation

- Which parametric model lies inside the band, and where does each fail? Can the drift be distinguished from zero? Does the answer depend on h , and what do the bands ignore?

C1: Which Continuous-Time Model Fits BET and Bitcoin? [Proposed]

Solution discussed in class

Task

- Data: BET (data/market/BET.csv, 2000–2026, $\Delta t = 1/252$) and Bitcoin (data/market/BTC-USD.CC.csv, 2014–2026, $\Delta t = 1/365$) daily log returns; no implied-volatility index covering the full sample is available for these markets
- (a) Fit GBM (B2), Merton (B6) and Heston by the CIR regression of B8 with the proxy $RV_t = \frac{1}{21\Delta t} \sum_{j=0}^{20} r_{t-j}^2$ in place of v_t
- (b) Simulate 100 samples of the same length from each fitted model and compute excess kurtosis, the Hill statistic of the largest losses ($k = 5\%$ of all returns), skewness and the sum of the ACF of $|r_t|$ over lags 1–20
- (c) For each model and statistic, report the percentile q of the data value in the simulated distribution: a descriptive check (parameters from the same data, not refitted on the simulated samples)
- (d) Open extension, with no reference solution: fit Bates (Heston plus Merton jumps) for latent volatility (simulated moments, particle filter) and repeat (b)–(c)
- Report: medians with 90% ranges and the table of percentiles q

Interpretation

- Which features does each model reproduce, and which none? Is Bitcoin a distinct class [Pele et al. \(2023\)](#)?

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "Assume the S&P 500 follows a GBM with $\mu = 8\%$ and $\sigma = 20\%$ per year. What is the probability that it ends the year below today's level? Check it by simulation."

The AI answer: text

- GBM: $dS = \mu S dt + \sigma S dW$
- By the chain rule, $d \ln S = dS/S = \mu dt + \sigma dW$, so $\ln(S_T/S_0) \sim N(\mu T, \sigma^2 T)$
- $P(S_T < S_0) = \Phi(-\mu\sqrt{T}/\sigma) = \Phi(-0.40) = 0.345$, with Φ the distribution function of the standard Normal distribution
- $\mathbb{E}[S_T] = S_0 e^{\mu T}$: on average the index ends the year 8.3% higher
- The simulation in the code gives 0.345 and confirms the formula

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np
mu, sigma, T, n, N = 0.08, 0.20, 1.0, 252, 20_000
dt = T / n
rng = np.random.default_rng(1)
Z = rng.standard_normal((N, n))
lnS = np.cumsum((mu - 0.5*sigma**2)*dt + sigma*dt*Z,
                axis=1)
print((lnS[:, -1] < 0).mean())
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ The answer contains three errors; everything else is correct: find them
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ Apply Itô's lemma to $f(S) = \ln S$ (A1): which term does the ordinary chain rule miss?
- ▣ Brownian increments over Δt have variance Δt : what is their standard deviation?
- ▣ Run the code: what does it actually print?

C3: Is S&P 500 Volatility Rough? [Proposed]

Solution discussed in class

Task

- ▣ Context: lecture case study [Gatheral, Jaisson & Rosenbaum \(2018\)](#) (arXiv version), Section 2 and eq. (5.1)
- ▣ Model: B1 (log-log slope with bootstrap CI), B4 (autoregression by OLS)
- ▣ Data: SPY 5-minute bars (data/market/intraday/SPY.US_5m.csv, 2020–2026), 09:30–16:00 New York time; Bitcoin (BTC-USD.CC_5m.csv, 2024–2026), 24-hour UTC days
- ▣ (a) For SPY and Bitcoin, compute the daily realised variance RV_t as the sum of squared 5-minute log returns within each day, and set $\sigma_t = \sqrt{RV_t}$
- ▣ (b) Compute $m(q, \Delta)$ for $q \in \{0.5, 1, 1.5, 2, 3\}$ and $\Delta = 1, \dots, 30$ days (overlapping increments); regress $\log m$ on $\log \Delta$; report ζ_q and ζ_q/q with a 90% moving-block bootstrap CI ([Künsch \(1989\)](#), blocks of 63 days)
- ▣ (c) For SPY, re-estimate $H = \zeta_2/2$ on the two halves of the sample, as in Table B.2
- ▣ (d) For SPY, forecast $\log RV_{t+\Delta}$ for $\Delta = 1, 5, 20$ with AR(5) and AR(10) (Yule–Walker) and HAR [Corsi \(2009\)](#) on a rolling 500-day window, and with the RFSV predictor (a Riemann sum of eq. (5.1)); compute P (Section 5.1)
- ▣ Report: ζ_q/q with CIs (against Table B.1), H by half, P (against Table 5.1), code and a two-page note

Interpretation

- ▣ Is SPY volatility rough, with a single exponent for every q ? Does the lead of RFSV over HAR grow with the horizon? Is Bitcoin rougher?

Common Pitfalls

- ▣ Forgetting the $-\sigma^2/2$ term: confusing the drift of the price with the drift of the log price
- ▣ Reporting a drift estimate without its standard error $\sigma/\sqrt{T}$
- ▣ Mixing units: annual parameters with daily Δt , trading days with calendar days (Bitcoin)
- ▣ Reading $\hat{\kappa}$ from a short sample as fast mean reversion: it is biased upwards
- ▣ Using χ^2 critical values for tests with a parameter on the boundary
- ▣ Treating $(\text{VIX}/100)^2$ as v_t : it is affine in v_t with risk-neutral parameters, so ξ is attenuated and the level contains a risk premium
- ▣ Hessian standard errors from a Gaussian quasi-likelihood: use sandwich standard errors, HAC when the scores are serially correlated

Take-Home Assignment (Team Project)

- ▣ Choose one asset not covered in class (a BVB stock, a currency or a crypto-asset)
- ▣ Fit GBM, Merton and a stochastic-volatility model; report every parameter with a standard error or a bootstrap CI
- ▣ Test jumps with Lee–Mykland and compare the jump count with the Merton intensity
- ▣ Simulate each model and rank them by how well they reproduce four stylised facts
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Applied Computational Finance
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

References (1/2)

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